

Digital Twins for Predictive Maintenance

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Abstract—**TODO:** Abstract here.

Index Terms—Digital Twin, Predictive Maintenance, Industry 4.0, Machine Learning, IoT, Condition Monitoring

ERKLÄRUNG ZUR NUTZUNG VON KI-TOOLS

TODO: KI-Erklärung hier einfügen (gemäß DHBW-Vorgabe).

I. INTRODUCTION

TODO: Introduction.

II. FUNDAMENTALS OF DIGITAL TWINS

TODO: Fundamentals of Digital Twins.

III. PREDICTIVE MAINTENANCE: CONCEPTS AND APPROACHES

TODO: Predictive Maintenance concepts.

IV. DIGITAL TWINS ENABLING PREDICTIVE MAINTENANCE

TODO: Integration of Digital Twin and PdM.

V. CASE STUDIES

TODO: Case study / case studies.

VI. ECONOMIC AND ORGANIZATIONAL ASPECTS

TODO: Economic and organizational aspects.

VII. CONCLUSION

TODO: Conclusion. [1]

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REFERENCES

- [1] M. Grieves, "Digital twin: Manufacturing excellence through virtual factory replication," *White Paper*, 2014.